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**Problem Set** 1 : Biophysics of solutions

**Exercise 1:**

1) What is the minimum mass of water ($g$) that must be introduced into a 5-liter chamber at a temperature of $37°C$ to obtain saturating water vapor?

2) The 5-liter chamber is compressed to a volume of 3 liters and $0.219g$ of water is then introduced into it. What would the vapor pressure be?

3) Same question as in 2) but taking the final volume of the chamber to be $V = 10l$.

Given for water at $37°C$ : $P_{sat} = 6.27kPa$

**Exercise 2:**

Consider, at temperature $T$ a solute $A$ dissolved in two immiscible fluids (1) and (2) . Let $\mu_1^0$ denote the standard chemical potential of $A$ in solvent (1) and $\mu_2^0$ the standard chemical potential of $A$ in solvent (2) . The two solutions are assumed to be ideal.

1) What is the relationship at equilibrium between the chemical potential $\mu_1$ of $A$ in (1) and the chemical potential $\mu_2$ of $A$ in (2)?

2) Deduce that, at equilibrium, the concentrations $C_1$ and $C_2$ of $A$ in (1) and (2) are related by $\frac{C_2}{C_1} = k$, where $k$ is a constant to be determined that depends only on temperature.

3) In a container, $1l$ of water (solvent (1)) and $1l$ of benzene (solvent (2)), two immiscible liquids, are poured. Then, $1mmole$ of phenol (solute $A$) is introduced into the container. The phenol dissolves in both water and benzene. Determine $C_1$ and $C_2$. Given: $T = 298K$; $\mu_1^0 - \mu_2^0 = -898J/mol$

$Resp : 0.7; 0.59mmol/l; 0.41mmol/l$

**Exercise 3:**

1 liter of solution contains:

$*$ $10ml$ of $HCl$ at $1mol/l$

$*$ $7.5ml$ of $H_2SO_4$ at $2mol/l$

$*$ $5.55g$ of $CaCl_2$ ($M = 111g/mole$)

$*$ $14.4g$ of glucose ($M = 180g/mole$)

The electrolytes are assumed to be fully dissociable. Calculate the osmolarity of the solution. $Resp : 295mOsm/l$

**Exercise 4:**

In one liter of urine with a freezing-point depression of $-1.4°C$, $11.7g$ of $NaCl$ and $15g$ of urea ($M = 60g/mol$) were found.

1) Calculate the total osmolarity, then those of $NaCl$ and urea.

2) What can we conclude from this? Can this excess osmolarity be attributed to albumin ($M_{al} = 70000g/mol$)? What other substance should be looked for?

Given for water: $K_{cr} = 1.86°C.Kg/osmol$

$Resp : 752mOsm/l; 393mOsm/l; 250mOsm/l$

**Exercise 5:**

$4.6g$ of ethyl alcohol ($M = 46g/mol$) in $200ml$ of pure solvent yield a solution that begins to deposit ice at $-0.93°C$.

Next, $5.8g$ of a non-ionizable compound whose molar mass is to be determined are dissolved in $100ml$ of the same solvent. The deposition of ice is found to begin at $-0.59°C$. Calculate the molar mass of the unknown substance.

Given: Freezing point of the pure solvent: $0°C$.

$Resp : 183g/mol$

**Exercise 6:**

Given that a $2.24M$ ammonia ($NH_4OH$) solution is iso-osmotic to a $1.4M$ sodium chloride solution, determine the dissociation coefficient $\alpha$ of $NH_4OH$. The membrane is considered semi-permeable.

$Resp : 0.25$